



CSE461

Section : 06

Group : 01

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Project report on real-world implications

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Smart Autonomous Vacuum & Wet Floor Mopping Robot

This project will design a small autonomous robot that will vacuum when the floor is dry, mop when the floor is wet, avoid obstacles, and accept sound-based commands. The goal will be to keep indoor spaces clean with minimal human effort while reducing risks linked to manual cleaning. The robot will use a moisture sensor to switch modes, ultrasonic sensors to navigate, and a sound sensor to respond to sounds like clap signals. By combining these features, the robot will serve as a practical, safe, and user-friendly solution that improves hygiene, supports vulnerable users, and demonstrates how robotics can make everyday life easier and more efficient.

Our smart cleaning robot will reduce the burden of repetitive vacuuming and mopping, allowing custodial staff in schools, offices, or malls to focus on tasks that will require human judgment and skill such as inspections or deep cleaning, which will raise the overall quality of cleanliness. While there will be concerns about job loss, the shift will more likely lead to job change, with new opportunities in robot setup, repair, and fleet supervision. The robot will also promote accessibility and inclusivity by helping elderly individuals, people with back pain, or those with limited mobility who will struggle with traditional cleaning, thereby restoring independence and making household chores more manageable in shared spaces. Culturally, it will challenge norms where cleaning is seen as women's work, turning it into a shared, technology supported responsibility, while in cultures that value ritual cleanliness it will help maintain those standards with less effort. We hope our robot will gain public trust and acceptance based on its ability to work quietly, safely, and efficiently, avoiding people and pets, operating at practical times, and giving clear signals of its actions, which will build confidence and reduce confusion or fear.

Our robot will improve hygiene by regularly vacuuming to reduce dust, dirt, and allergens, while wet mopping with proper solutions will lower microbial presence on hard floors, ensuring cleaner environments in classrooms, clinics, cafeterias, washrooms, and labs. It will also contribute to safer work environments, especially in hospitals, laboratories, or factories where hazardous spills of chemicals, bio fluids, or oils may occur, as the robot will quickly respond to such risks, reducing human exposure and protecting workers from long-term health issues. For

vulnerable users such as the elderly or individuals with back pain, arthritis, or balance problems, the robot will provide a practical way to maintain cleanliness without physical strain, and sound-triggered control will further support those who cannot operate small buttons or mobile applications. Additionally, by instantly cleaning food spills and damp patches, the robot will prevent the growth of insects and rodents, thereby reducing pest-related hazards and the spread of secondary diseases, making indoor environments healthier and safer for all.

The safety implications of our robot will be centered on preventing accidents and ensuring safe interaction between the machine and its environment. Slip and trip risks will be reduced because the moisture sensor will ensure mopping happens only when needed, while obstacle avoidance will help it steer around people, furniture, and cables. Human robot interaction will be made safer through simple and distinct sound cues, an emergency stop button, limited speed, and smooth movement, all of which will minimize risk during use.

The legal implications of our smart cleaning robot will confirm that it will fully comply with established robotic laws and safety standards. First, it will be designed never to harm humans, as its obstacle-avoidance system, moisture-based mode switching, low speed, and safe movement patterns will ensure that it will not cause injury, nor will it ignore risks that could indirectly cause harm through inaction. Second, it will be programmed to obey human commands, including sound-based inputs and an emergency stop, unless following such orders will endanger people, thereby aligning with the hierarchy of human safety. Third, the robot will be built to protect itself with sensors, protective casing, and controlled navigation, but it will always prioritize human safety and obedience over self-preservation when conflicts arise. Finally, it will include a clearly accessible and reliable kill switch, guaranteeing that users will be able to instantly disable it in emergencies, which will satisfy the legal requirement for ultimate human control. Thus, the robot will operate fully within the framework of robotic laws while ensuring safety, accountability, and user trust.

Cultural implications depend on the interaction of the technology with long-standing social values and traditions of cleaning. In the majority of cultures, cleaning has been regarded as a female activity, but our robots will break this stereotype by making cleaning an activity shared

by both sexes with the help of technology, thereby diminishing gender-based divisions of labor. In societies where ritual or religious cleanliness is highly prized, the robot will support those customs by keeping surfaces cleaner more often and with less human labor, allowing people and institutions to maintain traditions of cleanliness and order. The robot will also help to meet cultural changes of convenience and efficiency in the modern urban lifestyle where time will remain a scarce commodity, and it will take its place in the home and workplace as a symbol of progress. Cleaning robots will not only help to keep places clean but will also transform the way society perceives work, technology, and equality.

In conclusion, our Smart Autonomous Vacuum & Wet Floor Mopping Robot will represent a practical, safe, and inclusive solution for modern cleaning challenges by combining vacuuming, mopping, obstacle avoidance, and sound-based control in a single system. It will reduce human effort, improve hygiene, prevent accidents, and support vulnerable users while addressing societal, cultural, health, safety, and legal implications responsibly. By aligning with robotic laws, maintaining user control through a kill switch, and promoting equality in shared household and institutional tasks, the robot will not only enhance cleanliness and safety but will also reflect broader values of progress, accessibility, and responsible technology use. Ultimately, it will demonstrate how robotics will transform everyday life by making essential work more efficient, equitable, and sustainable.